

Lesson 2 · Homework

NCEA Level 2 · AS 91262 · Turning points and optimisation

Time: about 60 minutes

Name: _____

Date: _____

Show all working. Justify every maximum or minimum. [A] Achieved, [M] Merit, [E] Excellence.

1. Find the turning point of $y = x^2 + 6x - 7$ and state whether it is a maximum or a minimum. [A]

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2. Find the turning point of $y = 3 + 10x - x^2$ and state its nature. [A]

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3. Find the turning points of $f(x) = x^3 - 3x + 2$ and justify their nature. [A]

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4. Find the turning points of $f(x) = 2x^3 + 3x^2 - 36x + 1$ and justify their nature. [A]

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5. Find the three turning points of $y = x^4 - 8x^2$ and state their nature. [A]

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6. For which values of x is $f(x) = x^2 - 4x + 9$ decreasing? [A]

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7. For which values of x is $f(x) = x^3 - 3x^2 + 5$ increasing? [A]

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8. For which values of x is $f(x) = \frac{1}{3}x^3 - x^2 - 8x$ decreasing? [A]

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9. The curve $y = x^2 + bx + 5$ has a turning point at $x = 3$. Find b and the y -coordinate of the turning point. [M]

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10. A model rocket's height is $h = 40t - 5t^2$ metres after t seconds. Find its maximum height. [A]

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11. The daily cost of making x units is $C = 2x^2 - 80x + 1000$ dollars. Find the number of units that gives the minimum cost, and the minimum cost. [A]

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12. Two positive numbers add to 20. Use calculus to find the largest possible value of their product. [M]

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13. A rectangular garden against a house wall is fenced on the other three sides with 40 m of fencing. Find the dimensions that give the largest area, and that area. [M]

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14. **Extension** A rectangle is drawn under the curve $y = 12 - x^2$ with two corners on the x -axis and two corners on the curve, symmetrical about the y -axis. Find the largest possible area of the rectangle, and justify that it is a maximum. [E]

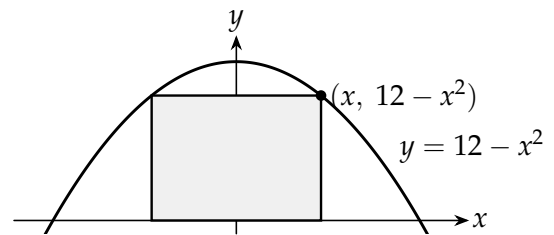
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Optional extra: StudyTime Level 2 Calculus Guide, "Calculating Turning Points" from page 16.